



Enhanced mechanical properties of HfMoTaTiZr and HfMoNbTaTiZr refractory high-entropy alloys



Chien-Chang Juan^a, Ming-Hung Tsai^b, Che-Wei Tsai^a, Chun-Ming Lin^a,
Woei-Ren Wang^c, Chih-Chao Yang^c, Swe-Kai Chen^d, Su-Jien Lin^a, Jien-Wei Yeh^{a,*}

^a Department of Materials Science and Engineering, National Tsing Hua University, Hsinchu 30013, Taiwan

^b Department of Materials Science and Engineering, National Chung Hsing University, Taichung 40227, Taiwan

^c Industrial Technology Research Institute, Tainan 70955, Taiwan

^d Center for Nanotechnology, Materials Science, and Microsystems, National Tsing Hua University, Hsinchu 30013, Taiwan

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ABSTRACT

Although refractory high-entropy alloys have exceptional strength at high temperatures, they are often brittle at room temperature. One exception is the HfNbTaTiZr alloy, which has a plasticity of over 50% at room temperature. However, the strength of HfNbTaTiZr at high temperature is insufficient. In this study, the composition of HfNbTaTiZr is modified with an aim to improve its strength at high temperature, while retaining reasonable toughness at room temperature. Two new alloys with simple BCC structure, HfMoTaTiZr and HfMoNbTaTiZr, were designed and synthesized. The results show that the yield strengths of the new alloys are apparently higher than that of HfNbTaTiZr. Moreover, a fracture strain of 12% is successfully retained in the HfMoNbTaTiZr alloy at room temperature.

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1. Introduction

In recent years there has been a pressing need to increase the efficiency of gas turbine engines in the aerospace industry. This is mainly imposed by the reduction of fuel consumption and greenhouse gas emission. One of the most effective methods to increase engine efficiency is to increase the operating temperature of the engine. The prevailing engine component material is Ni-based superalloys. Unfortunately, the incipient melting point of Ni clearly sets a limitation to the operation temperature. Therefore, in developing the next-generation gas turbines, a sufficiently high melting point becomes critical. Refractory metals, such as Nb and Mo, have high melting points and high solubility for solid-solution-strengthening elements, and therefore are considered as possible candidates for such applications. For example, Nb–Si alloys have a good combination of high-temperature strength and low-temperature damage tolerance [1,2]. Mo–Si–B alloys exhibit outstanding creep strength and high-temperature strength [3–5].

A new alloy design strategy called high-entropy alloys (HEAs) has drawn significant attention in the past decade [6–9]. These alloys can have high hardness, good oxidation and corrosion resistance, good structural stability, and excellent high temperature strength [10–24]. These advantages make HEAs particularly attractive for high-temperature applications. The combination of the above lines of thinking gives good reasons to synthesize HEAs using refractory elements – or refractory HEAs – as proposed by Senkov et al. [24–29]. Refractory HEAs show very encouraging properties. For example, the MoNbTaVW alloy has a remarkably high yield strength of 477 MPa at 1600 °C [24]. However, the main problem of this alloy is its low toughness (fracture strain = 1.7%) at room temperature (RT) and its high density 12.36 g/cm³. Another refractory HEA, HfNbTaTiZr, having very different properties was developed later [26]. It has a lower density (9.94 g/cm³) and exhibits high compression yield strength ($\sigma_{0.2}$ = 929 MPa) and very high plasticity (ϵ > 50%) at RT, but its yield strength at high temperatures is not pleasing (yield strength < 100 MPa at 1200 °C). In this study, two new alloys are designed by modifying the composition of the HfNbTaTiZr alloy. The first is HfMoTaTiZr, obtained by replacing Nb with Mo. The improvement in mechanical properties is expected from the report that the addition of Mo could improve

* Corresponding author. Tel.: +886 3 5719558.

E-mail address: jwyeh@mx.nthu.edu.tw (J.-W. Yeh).

high temperature mechanical properties of Nb-xMo-22Ti–18Si in situ composites [30] and also from the previous research [31] that the hot hardness of AlCoCrFeMo_{0.9}Ni alloys increases with the increased addition of Mo (see Fig. 1). In addition, the present base alloy, HfNbTaTiZr, having single BCC phase has a large solution hardening effect [26]. Thus, Mo having a triple Young's modulus and a higher melting point by around 200 °C is expected to potentially increase both hardness and strength of the base alloy according to solution hardening mechanism. Although the replacement of Nb by heavier Mo could increase the alloy density by ~2.7%, the improvement of the high-temperature strength is expected to be larger to recoup this small disadvantage. The second is HfMoNbTaTiZr, which is obtained by adding Mo to the base alloy HfNbTaTiZr. This is intended to see the effect of Mo and also the effect of Nb as compared with the first alloy HfMoTaTiZr. In brief, our goal is to improve the high-temperature strength of HfNbTaTiZr while retain a reasonable plasticity at RT. In addition, various characterizations of these two alloys are presented and discussed with further insights via physical metallurgy approach. Another purpose of this research is to give a basis for further alloy and process design in achieving even better properties.

2. Experimental procedures

The two refractory HEAs, HfMoTaTiZr and HfMoNbTaTiZr, were prepared by vacuum arc melting in an argon atmosphere. The raw materials had purities higher than 99.9 wt.%. Melting was repeated for at least 5 times to improve the chemical homogeneity of the alloy. Crystal structures of the alloys were examined with an X-ray diffractometer (Shimadzu XRD-6000) operated at 30 kV and 20 mA. The scanning rate is 4°/min. Vickers hardness was measured by Mitutoyo HV-115 hardness testing machine with a load of 5 kg. The duration time in each indentation was 15 s. Compression tests at RT, 800, 1000, and 1200 °C were carried out using a Gleeble-3500 thermal-mechanical simulator. Tests were conducted under argon atmosphere after purging with argon for three times. Cylindrical samples for compression test were 3 mm in diameter and 6 mm in height. The samples were cut by wire electrical discharge machining and with the sample axis perpendicular to the bottom surface of the button-shaped ingot. This is intended to let the solidification direction and thus columnar zone approximately be parallel to the sample axis. In the test, the samples were resistance heated to testing temperature at a heating rate of 20 °C/s by thermocouple feedback-controlled AC current and then holding isothermally for 180s before deformation was initiated. The samples were tested at an initial strain rate of 10^{−3}/s. Microstructures of

as-solidified and deformed samples were analyzed by using scanning electron microscopes (SEM, JEOL JSM-5410 and JSM-6390). The chemical compositions of these as-solidified alloys were analyzed by SEM energy dispersive spectrometry (EDS). The precipitates in deformed samples were examined with transmission electron microscopy (TEM, JEOL JEM-2010).

3. Results and discussion

3.1. Crystal structure, lattice constant, density and hardness

Fig. 1 shows the X-ray diffraction patterns of as-solidified HfMoTaTiZr and HfMoNbTaTiZr alloys. The crystal structures of both alloys are identified to be BCC structure (cI2-W). The shape of the (200) and (211) peaks are asymmetric. As will be shown later, this is a result of the compositional difference between dendrite and interdendrite regions, which leads to a slight difference in the lattice constant between dendrite and interdendrite regions. The lattice constants of the HfMoTaTiZr and HfMoNbTaTiZr alloys were determined from the XRD peaks to be $a_{\text{HfMoTaTiZr}} = 3.376$ Å and $a_{\text{HfMoNbTaTiZr}} = 3.370$ Å, respectively. These values are compared with those calculated by rule-of-mixture (ROM) or Vegard's law [32]:

$$a_{\text{mix}} = \sum c_i a_i$$

where c_i and a_i are the atomic fraction and lattice parameter of the i th element, respectively. The lattice constants of all related pure elements (i.e., a_i) are listed in Table 1. The calculated lattice constants of the HfMoTaTiZr and HfMoNbTaTiZr alloys are 3.373 and 3.361 Å, respectively. These values agree well with those obtained experimentally. The theoretical density of a solid-solution alloy is estimated by the equation:

$$\rho_{\text{mix}} = \frac{\sum c_i A_i}{\sum c_i A_i / \rho_i}$$

where A_i and ρ_i are the atomic weight and density of element i , respectively. The theoretical densities of the HfMoTaTiZr and HfMoNbTaTiZr alloys are 10.21 g/cm³ and 9.95 g/cm³, respectively. These values are also close to the experimental values, $d_{\text{HfMoTaTiZr}} = 10.24$ g/cm³ and $d_{\text{HfMoNbTaTiZr}} = 9.97$ g/cm³.

The Vickers hardness of the HfMoTaTiZr and HfMoNbTaTiZr alloys are Hv 542 and Hv 505, respectively. In stark contrast to the cases of lattice constant and density, the measured hardness values are about three times higher than that calculated by the rule of mixture, i.e., Hv 123 and Hv 125. Such significant hardening in multi-principal-element BCC solid solutions has been observed in many refractory HEAs, and has been attributed to solution hardening mechanism [25,26,33–35].

3.2. Dendritic structures

Fig. 2 shows the SEM backscattered electron images of as-solidified HfMoTaTiZr and HfMoNbTaTiZr alloys, which are taken from the cross-section around 2 mm distance from the bottom surface of the solidified ingots. Both alloys show typical cast dendritic structure. The chemical compositions of dendrite and interdendrite regions are listed in Table 2. The dendrite of the HfMoTaTiZr alloy is rich in high-melting-point elements: Ta (2996 °C) and Mo (2610 °C). In contrast, the interdendrite region is rich in low-melting-point elements: Hf (2222 °C), Ti (1668 °C), and Zr (1852 °C). Similarly, the dendrite region of the HfMoNbTaTiZr alloy is rich in Ta, Mo and Nb (2415 °C). This is consistent with previous reports on refractory HEAs and the reason has been

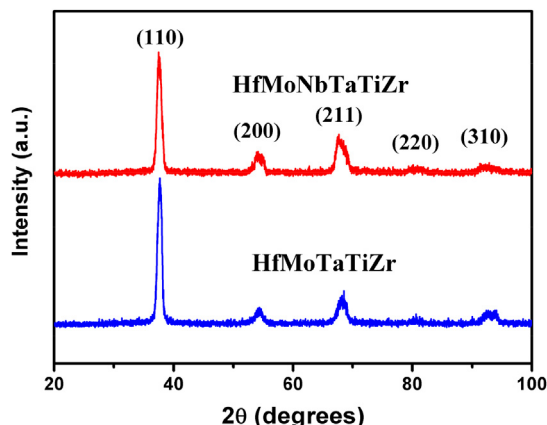


Fig. 1. XRD patterns of as-solidified HfMoTaTiZr and HfMoNbTaTiZr alloys.

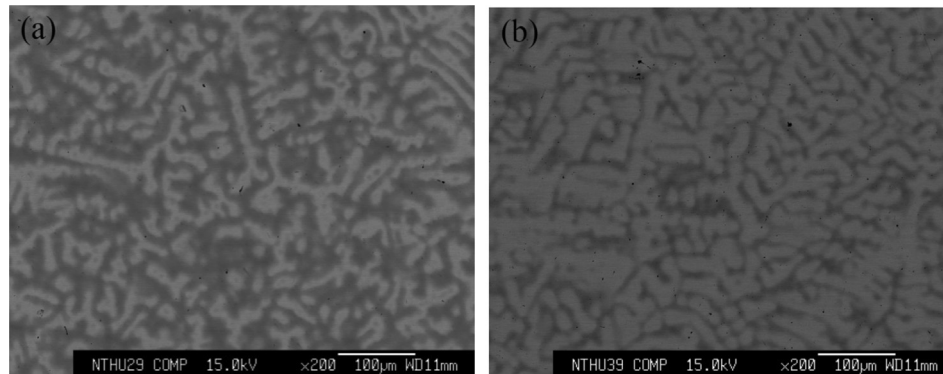
Table 1

Melting point, lattice constant, density, and Vickers hardness of the two HEAs and their composing elements.

	Hf	Mo	Nb	Ta	Ti	Zr	HfMoTaTiZr	HfMoNbTaTiZr
Melting point (°C)	2222	2610	2415	2996	1668	1852	—	—
Lattice constant (Å)	3.559 ^a	3.147	3.301	3.303	3.276 ^a	3.582 ^a	cal: 3.373 exp: 3.376	cal: 3.361 exp: 3.370
Density ^b (g/cm ³)	13.15	10.22	8.58	16.68	4.52	6.59	cal: 10.21 exp: 10.24	cal: 9.95 exp: 9.97
Vickers hardness (kg/mm ²)	180	156	135	89	99	92	cal: 123 exp: 542	cal: 125 exp: 505

^a The lattice constants of Ti, Zr, and Hf are extrapolated from their respective high temperature BCC phase to RT (the structures of Ti, Zr, and Hf are BCC at high temperatures) using their coefficients of thermal expansion [26].

^b The densities of all pure elements are calculated based on the lattice constants of their BCC structure.

**Fig. 2.** SEM backscattered electron images of as-solidified (a) HfMoTaTiZr and (b) HfMoNbTaTiZr alloys.

attributable to the strong tendency for elements with higher melting point to form the first solid phase [24–26].

3.3. Strength, fracture strain and fracture feature from 25 to 1200 °C

The engineering stress–strain curves of HfMoTaTiZr and HfMoNbTaTiZr obtained by compression test at various temperatures are shown in Fig. 3. The yield strength and fracture strain are taken from Fig. 3 and summarized in Table 3. The yield strength of the HfMoTaTiZr alloy at RT is 1600 MPa. The alloy shows its peak strength of 1743 MPa at $\epsilon = 3\%$, and subsequently fails at $\epsilon = 4\%$. The yield strength of the HfMoNbTaTiZr alloy at RT is 1512 MPa. The peak strength of this alloy occurring at $\epsilon = 11\%$ is 1828 MPa. The maximum compression strain is 12%. Thus, HfMoNbTaTiZr alloy exhibits a significantly higher plasticity than HfMoTaTiZr alloy does. Instead of inclining at $\sim 45^\circ$ to the compression direction, the fracture surfaces of both alloys are parallel to the compression direction. This suggests that the lateral tensile stress induced by longitudinal loading is the cause for the mode-I fracture along longitudinal direction. The HfMoNbTaTiZr alloy exhibits a significantly higher plasticity than the HfMoTaTiZr alloy does. Fig. 4 shows the fracture surfaces of the two alloys. The fracture surfaces of the

two alloys are similar and contain both cleavage and quasi-cleavage features. Cleavage fracture is featured by either flat surface or river pattern whereas quasi-cleavage is featured by dense slip lines on the cleavage surface. Compared with the fracture surface of HfMoTaTiZr, less cleavage features are observed on the surface of HfMoNbTaTiZr. In fact, in some areas features of dimpled-rupture are observed (Fig. 4(c)). This means that HfMoNbTaTiZr alloy have larger shear deformation and thus higher resistance to fracture than HfMoTaTiZr alloy. Because the main difference between the two alloys is the addition of Nb, it suggests that Nb can increase the plasticity of refractory HEAs. However, the mechanism behind the alloying effect of Nb is not discussed here.

The high temperature mechanical properties of the two alloys have been investigated, also by compression test, at 800, 1000, and 1200 °C. For the HfMoTaTiZr alloy, the yield strength at 800 °C is 1045 MPa and the maximum strength occurring at $\epsilon = 16\%$ is 1446 MPa. Obvious strain hardening is observed. The yield strengths at 1000 and 1200 °C are 855 and 404 MPa, respectively. At these temperatures, the stress after yielding gradually descends until the test stops at $\epsilon = 30\%$ without fracture. The softening indicates that rate of dislocation accumulation is not sufficient to harden the alloy. Also, soft phase formed in highly deformed regions, as will be seen later.

Table 2

Chemical compositions (in at.%) in the HfMoTaTiZr and HfMoNbTaTiZr alloys.

Alloy		Hf	Mo	Nb	Ta	Ti	Zr
HfMoTaTiZr	Nominal	20.0	20.0	—	20.0	20.0	20.0
	Dendrite	14.9	23.8	—	32.1	18.3	10.9
	Interdendrite	22.4	15.5	—	8.2	20.1	33.8
HfMoNbTaTiZr	Nominal	16.7	16.7	16.7	16.7	16.7	16.7
	Dendrite	14.3	18.4	19.5	24.4	12.4	10.8
	Interdendrite	21.1	13.6	12.3	9.9	18.3	24.7

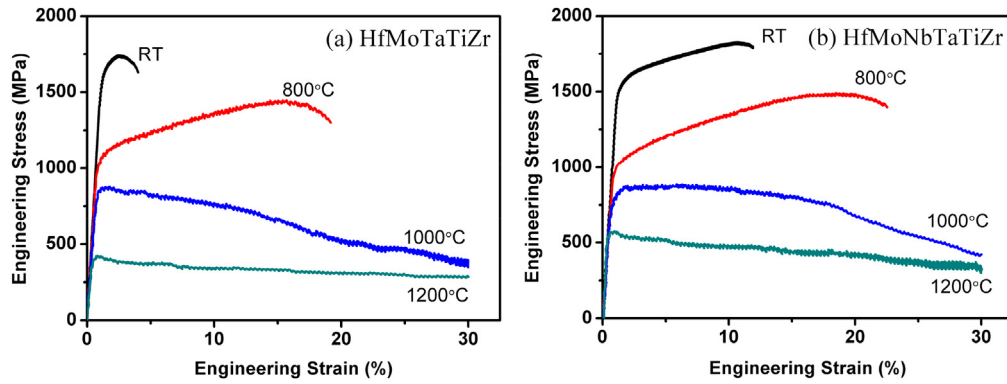


Fig. 3. Engineering stress–strain curves of (a) HfMoTaTiZr and (b) HfMoNbTaTiZr alloys obtained by compression test.

Table 3

The yield strength ($\sigma_{0.2}$) and fracture strain (ϵ_f) of the HfMoTaTiZr and HfMoNbTaTiZr alloys at different temperatures.

Test Temperature (°C)	HfMoTaTiZr		HfMoNbTaTiZr	
	Yield strength $\sigma_{0.2}$ (MPa)	Fracture strain ϵ_f (%)	Yield strength $\sigma_{0.2}$ (MPa)	Fracture strain ϵ_f (%)
25	1600	4	1512	12
800	1045	19	1007	23
1000	855	>30	814	>30
1200	404	>30	556	>30

For the HfMoNbTaTiZr alloy, yielding takes place at around 1007 MPa at 800 °C. The plastic strain is over 20%, and obvious strain hardening is observed. The maximum strength is 1489 MPa which occurs at $\epsilon = 19\%$. The yield strengths of HfMoNbTaTiZr at 1000 and 1200 °C are 814 and 556 MPa, respectively. Similar to the behavior of HfMoTaTiZr, the strength immediately reaches its maximum and then slowly decline until the test stops. At RT,

800 °C, and 1000 °C, the yield strengths of two alloys are similar. At 1200 °C, the yield strength of the HfMoNbTaTiZr alloy (556 MPa) is higher than that of the HfMoTaTiZr alloy (404 MPa). This suggests that the HfMoNbTaTiZr alloy has higher resistance to thermal softening. In terms of plasticity, the HfMoNbTaTiZr alloy has a significantly higher strain to failure than the HfMoTaTiZr alloy at RT. The difference in plasticity between the two alloys is still

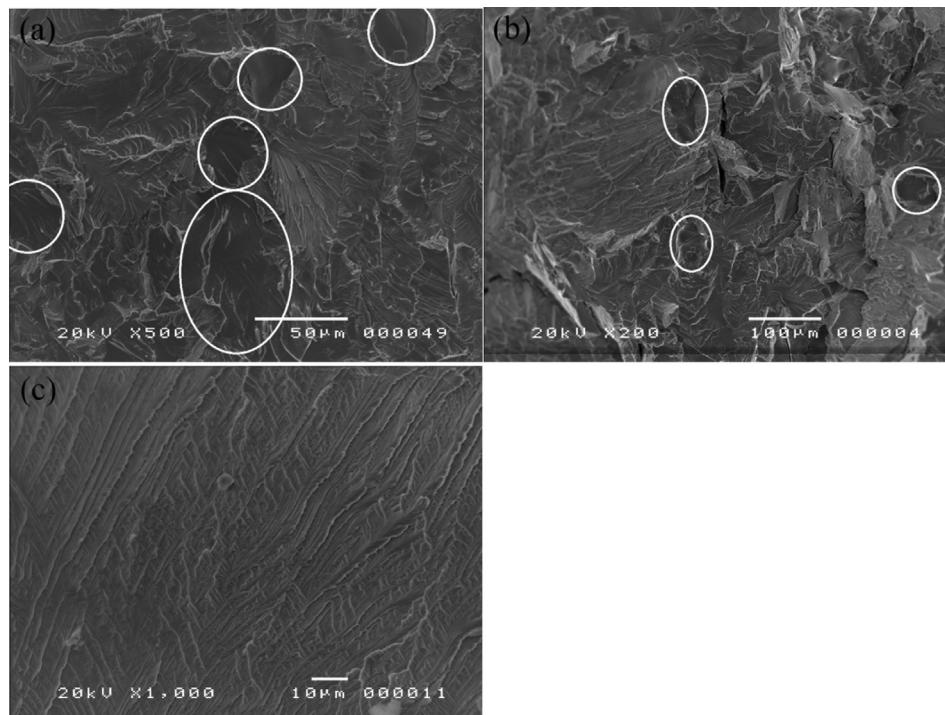


Fig. 4. SEM images of the fracture surfaces of (a) HfMoTaTiZr, (b) HfMoNbTaTiZr, and (c) HfMoNbTaTiZr alloys (high magnification) after compression deformation at 25 °C. The circles indicate the cleavage regions.

observable at 800 °C. At temperatures higher than 800 °C, however, the difference in plasticity is not distinguishable because the test stops at $\epsilon = 30\%$. By and large, the combination of strength and plasticity of the HfMoNbTaTiZr alloy is better than that of HfMoTaTiZr.

3.4. Strength comparison: base alloy and typical superalloys

Fig. 5(a) compares the strengths of the two modified alloys with that of base alloy, Inconel 718 [36] and CMSX-4 [37]. The compression tests of two modified alloys and the based alloy were at the same initial strain rate $10^{-3}/s$. However, the data of Inconel 718 and CMSX-4 were obtained by tension test at a strain rate of 0.005/min up to yield and then at another strain rate of 0.05/min after yielding. Compared with the base alloy, in the as-solidified condition (with the same processing for the modified alloys, also marked as this work) or in the hot-isostatic pressed and vacuum annealed condition [27], the strengths of two modified alloys are markedly higher. For example, the yield strengths of both alloys are higher than that of the base alloy across the whole temperature range covered in this study. At 1200 °C, their yield strengths (404 and 556 MPa) are 4.3 and 6 times that of the base alloy. The specific yield strength (i.e., yield strength to density ratio) shown in

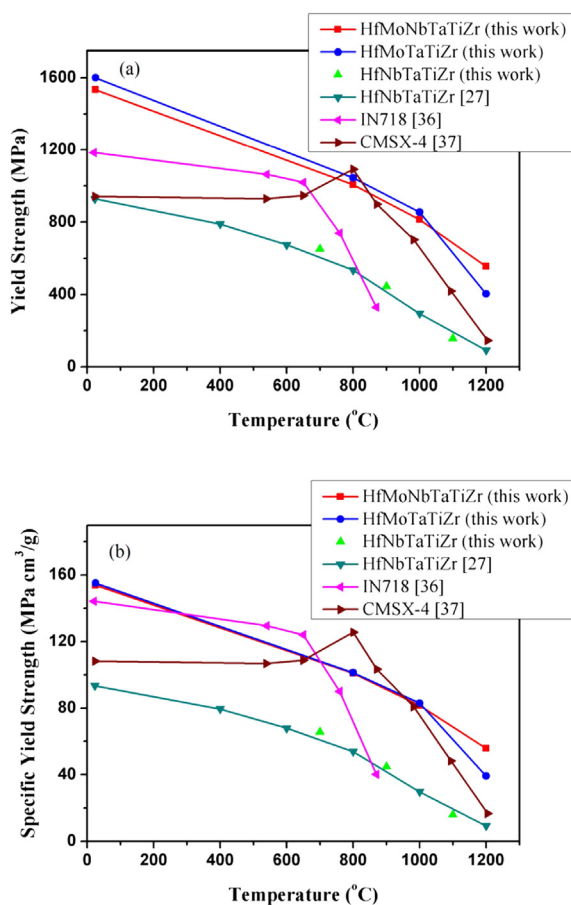


Fig. 5. Temperature dependence of (a) yield strength and (b) specific yield strength of HfMoTaTiZr, HfMoNbTaTiZr, HfNbTaTiZr (in as-solidified condition done by this work for comparison), HfNbTaTiZr (in the hot-isostatic pressed and vacuum annealed condition) [27] HEAs, and two superalloys, Inconel 718 [36] and CMSX-4 [37] (the yield strengths specific yield strengths and of Inconel 718 and CMSX-4 are tensile yield strengths). Nominal composition of Inconel 718 is Ni: 52.5%, Cr: 19.0%, Mo: 3.0%, Nb: 5.1%, Al: 0.5%, Ti: 0.9%, Fe: 18.5%, Mn: 0.2%, Si: 0.2%, and B: 0.04%. That of CMSX-4 is Cr: 6.5%, Co: 10.0%, Mo: 0.6%, W: 6.5%, Ta: 6.5%, Re: 3.0%, Al: 5.5% and Ni: As Balance (in wt.%).

Fig. 5(b) also show the same trend – the modified alloys apparently outperform the base alloy. Consequently, the addition of Mo is quite effective in strengthening the refractory HEAs. Compared with Inconel 718 and CMSX-4 (Fig. 5(a)), both modified alloys have higher yield strengths at all test temperatures except for CMSX-4 at 800 °C. Moreover, the resistances to thermal softening for both HEAs are higher than those of the superalloys. This is seen by comparing the transition temperature. According to the theory of Westbrook [38,39], there are two stages in strength–temperature curves. At the first stage (low temperature region), the softening is through dislocation slip. At the second stage, the softening mechanism is related to diffusional phenomenon such as (1) slip along more than one slip system due to increased atomic mobility, (2) dislocation climb, and (3) grain boundary rotation. Indeed, the transition temperatures between stage I and stage II for HfMoTaTiZr (~1000 °C) and HfMoNbTaTiZr (probably above 1200 °C) are markedly higher than those for CMSX-4 (800 °C) and Inconel 718 (700 °C). This indicates that addition of Mo apparently inhibit diffusional plastic behavior and enhances the resistance to thermal softening.

Due to the high strengths at high temperatures, the specific yield strengths of both modified alloys at 1000 and 1200 °C are also higher than those of the superalloys. Thus, both alloys have great potential in high-temperature applications over 1000 °C. Because of its better combination of strength and plasticity, HfMoNbTaTiZr is even better than HfMoTaTiZr. However, further studies on creep properties are needed for applications that require creep resistance.

3.5. Microstructure evolution during high-temperature deformation

Fig. 6 shows the longitudinal cross-section microstructure of HfMoTaTiZr alloy after compression test at different temperatures. The loading direction is the vertical axis. The observation was made on the central region of samples to avoid the constraint effect from outer cold zones. At 800 °C, (Ta, Mo)-rich dendrite (white regions) and (Zr, Hf)-rich interdendrite (gray regions) are still clearly visible after compression. Dendrites and interdendrites are generally observed to elongate laterally. In some regions, however, the dendritic structure are elongated in directions about 60° inclined to the loading direction, which are related to the formation of shear bands.

Compression at 1000 °C leads to significant deformation. Almost all grains are elongated laterally. In addition, high magnification image shown in Fig. 6(c) reveals the formation of Mo-rich phase (marked by white arrows) in the interdendrite region. The Mo-rich precipitates have sizes between 100 and 500 nm, and spread along the highly deformed regions due to the severe plastic flow of interdendrite regions. It is noted that these Mo-rich precipitates typically locate along the band which comes from the outer region of dendrites, that is, the transition zone between dendrite and interdendrite. Fig. 7 shows the TEM bright-field image of the highly deformed region after 1000 °C compression test. The TEM sample was taken from the specific region by focus ion beam (FIB) technique. Mo-rich and (Zr, Hf)-rich phases with sizes around 100 nm were discovered by TEM investigation and their compositions were analyzed by EDS (Fig. 7 and Table 4). This indicates that the phase transformation preferentially occurred in the highly deformed regions. A possible explanation of the phenomenon is that the precipitation is kinetically facilitated by the increased defect density in the highly sheared region. In addition to the BCC Mo-rich phase, the crystal structure of the (Zr, Hf)-rich phase is identified by two nano-beam electron diffraction to be simple face-centered cubic (FCC) with a lattice constant of 4.63 Å (Fig. 7). This result is quite intriguing because of two points. 1. The Hf–Zr binary phase

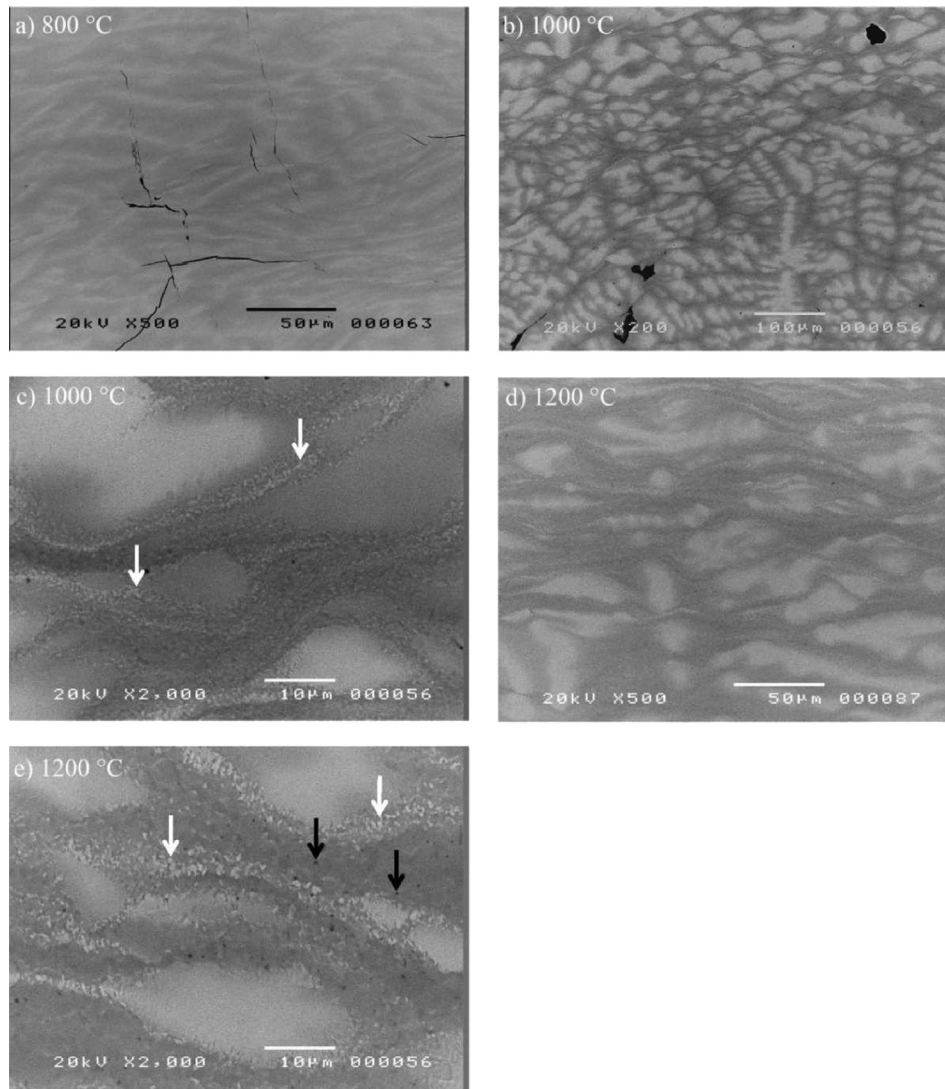


Fig. 6. SEM backscattered electron images of the microstructure of HfMoTaTiZr alloy after compression deformation at (a) 800 °C, (b) 1000 °C, (c) 1000 °C with higher magnification, (d) 1200 °C and (e) 1200 °C with higher magnification.

diagram suggests complete solubility between the two elements, and the crystal structure of this solid solution at low temperatures is HCP whereas that at higher temperature is BCC [40]. 2. The bulk alloy with the composition of (Zr, Hf)-rich phase was also synthesized and analyzed for comparison. The XRD result (not shown here) shows that bulk (Zr, Hf)-rich alloy has hexagonal crystal structure at room temperature. However, the reason why the (Zr, Hf)-rich phase crystallizes into FCC instead of HCP requires further study.

At 1200 °C, the grains are also severely deformed and elongated laterally. The Mo-rich phase and the (Zr, Hf)-rich phase are again formed in the highly sheared region. The Mo-rich phase (white particles) has a larger size, 500 nm in diameter, than that formed during 1000 °C compression. Besides, the (Zr, Hf)-rich phase (dark particles) also have a larger size around 350 nm in diameter. Here we recall the strain softening phenomenon observed at 1000 °C and 1200 °C. The phase transformation observed is believed to be the mechanism of softening. Once the Mo-rich and (Zr, Hf)-rich phases are formed, the effect of solution hardening is reduced.

The microstructures of the HfMoNbTaTiZr alloy after compression at different temperatures are shown in Fig. 8. The loading

direction is the vertical axis. After compression at 800 °C, some dendrite/interdendrites are elongated along directions inclined at ~55° to the compression direction. This is related to the formation of shear bands. Both vertical and inclined cracks are seen. Inclined cracks originate from the highly localized strain in the shear bands. For vertical cracks, the cause is the high lateral tensile stress induced by the compression. At 1000 °C, the grains experience more compressive deformation and some shear bands are also formed (marked by dash line). Besides, a small number of cavities are seen to form in the shear bands. Unlike the HfMoNbTaTiZr alloy, no second phases formed during deformation are found in HfMoNbTaTiZr at 1000 °C, at least by the observation of SEM. Therefore, the precipitation temperature is lower in HfMoNbTaTiZr.

Compression at 1200 °C leads to a significant amount of plastic flow, which squashes the cast dendritic structures. However, no cavities are discovered. Higher magnification images in Fig. 8(d) reveal the formation of (Zr, Hf)-rich (black contrast) and Mo-rich (white contrast) precipitates in highly deformed interdendrite region. The diameter of (Zr, Hf)-rich precipitates are 300 nm whereas that of Mo-rich ones are 160 nm, both of which are smaller than the precipitates in HfMoTaTiZr alloy at 1200 °C.

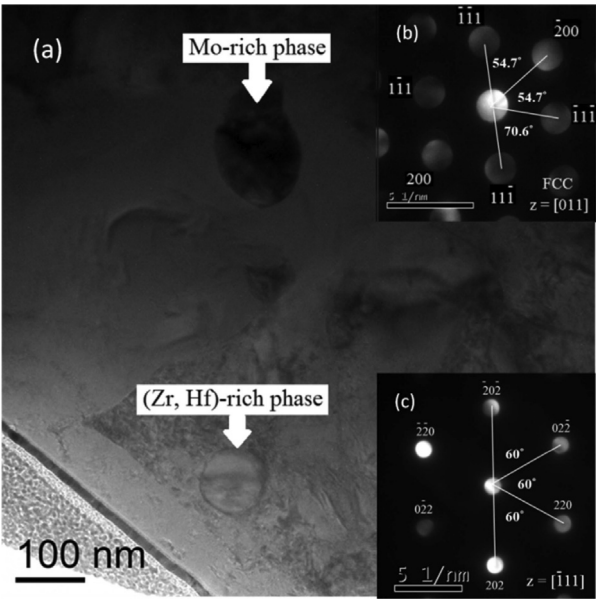


Fig. 7. (a) TEM image of the HfMoTaTiZr alloy after compression testing at 1000 °C; (b) and (c) nano-beam diffraction patterns of (Zr, Hf)-rich phase with FCC [011] and FCC [111] zone axes, respectively.

Table 4
Chemical composition (at.%) of the phases in the interdendrite region of the HfMoTaTiZr alloy after compression at 1000 °C.

	Hf	Mo	Ta	Ti	Zr
(Zr, Hf)-rich phase	28.5	3.7	5.6	7.0	55.2
Mo-rich phase	11.5	31.6	17.2	22.0	17.7

4. Conclusions

Two new refractory HEAs, HfMoTaTiZr and HfMoNbTaTiZr, were designed based on the HfNbTaTiZr alloy in order to improve the overall properties. Several findings are drawn:

- (1) Both alloys have simple BCC structure. The lattice constants and densities of the two alloys are close to the values obtained by rule-of-mixture. In contrast, the hardnesses of both alloys are about three times higher than the rule-of-mixture values. This suggests a strong solid-solution strengthening.
- (2) The compression yield strengths of the HfMoTaTiZr alloy at RT, 800, 1000, and 1200 °C are 1600, 1045, 855, and 339 MPa, respectively. The corresponding values for the HfMoNbTaTiZr alloy are 1512, 1007, 814, and 556 MPa, respectively.
- (3) The fracture strain of the HfMoTaTiZr alloy is 4% and that for the HfMoNbTaTiZr alloy is 12% at RT. The fracture surfaces of the two alloys suggest that the fracture mode is mixture of cleavage and quasi-cleavage. Less cleavage fracture features are observed in the HfMoNbTaTiZr alloy.
- (4) As temperature increases, the yield strengths of the alloys decrease while plastic strains increase. The combination of strength and plasticity for the HfMoNbTaTiZr alloy is better than that of HfMoTaTiZr at all test temperatures. During the compression at 1000 °C (for HfMoTaTiZr) or 1200 °C (for both alloys), Mo-rich and (Zr, Hf)-rich precipitates form in the interdendrite regions. The crystal structure of the (Zr, Hf)-rich precipitate is FCC, which is different from the prediction of binary phase diagram.
- (5) Compared with the base alloy (HfNbTaTiZr), the strengths of the two modified alloys at elevated temperatures are markedly higher. The yield strength of HfMoNbTaTiZr is more than six times that of HfNbTaTiZr at 1200 °C. The yield strength values are higher than that of Inconel 718 and CMSX-4 at all testing temperatures except for CMSX-4 at 800 °C. The

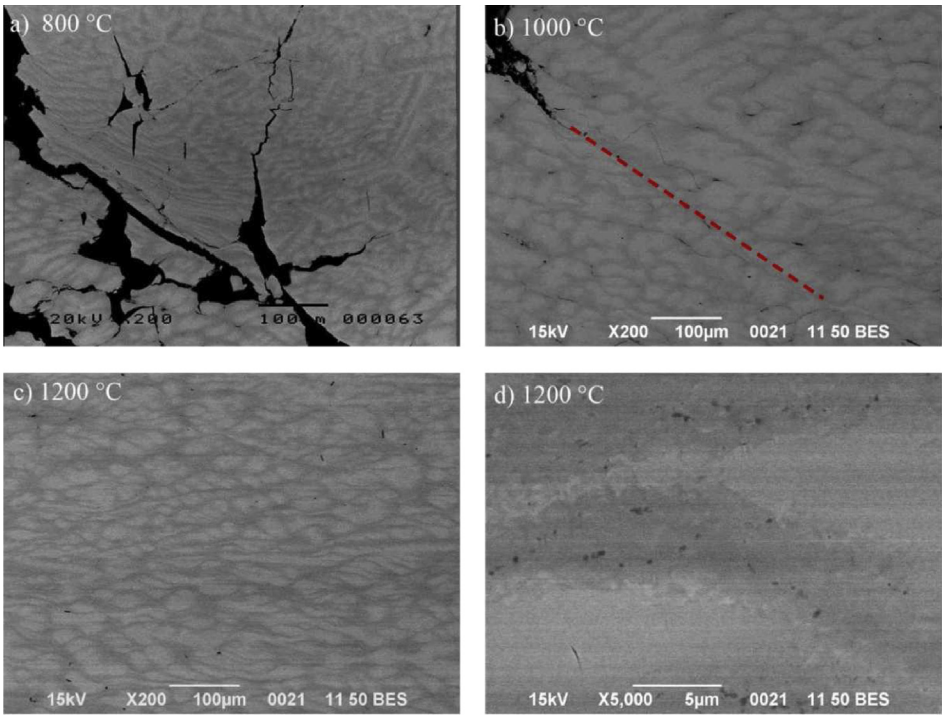


Fig. 8. SEM backscattered electron images of the microstructure of HfMoNbTaTiZr after compression deformation at (a) 800 °C, (b) 1000 °C, (c) 1200 °C, and (d) 1200 °C with higher magnification.

resistances to thermal softening for both modified alloys are also higher than those of the superalloys.

- (6) Both new alloys have great potential in high-temperature applications, yet the combination of strength and plasticity in HfMoNbTaTiZr is better than that in HfMoTaTiZr.

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